Clinical Treatment-Outcome Model Comparison & Reflective Report

Generated 2026-09-08 10:56. Dataset: 1,015,405 rows, 39 columns, target treatment_outcome.

1. Executive Summary

Six models (Decision Tree, Random Forest, XGBoost, Gradient Boosting, LSTM, MLP) were tuned and evaluated under a single leakage-safe 60/20/20 stratified split (random_state=42) to predict treatment_outcome (0=Ineffective, 1=Effective). Because Effective cases represent only 22.34% of the data, PR-AUC (average precision) was used as the primary tuning and comparison metric throughout, rather than accuracy. All six final models achieved PR-AUC between 0.3320 and 0.3465, well above the majority-class baseline of 0.2234 but clustered tightly together, indicating the dataset's feature set carries limited predictive signal (see Section 3). The selected champion model is **Gradient Boosting**.

2. Dataset Overview

Source: dataset/feature_engineered_clinical_data.csv — 1,015,405 rows, 39 columns, 2,461 missing cells. Features span demographics, vitals, lab values, medication/dosage details, and engineered clinical flags (kidney_stage, liver_risk, polypharmacy, bmi_category, age_group, elderly_high_dose, de_ritis_ratio). patient_id was dropped from modeling features. Preprocessing (median imputation for numeric columns, frequency-rank encoding for categoricals, and -- for the two neural networks only -- StandardScaler) was fit strictly on the training partition and applied unchanged to validation/test.

3. Target Imbalance and Why Accuracy Is Not the Primary Metric

Class 0 (Ineffective) = 788,599 (77.66%); Class 1 (Effective) = 226,806 (22.34%). A trivial classifier that always predicts 'Ineffective' achieves 77.66% accuracy while identifying zero Effective cases (Recall=0, Precision=0, PR-AUC≈0.2234, the class prevalence). This demonstrates why accuracy alone is misleading here: every model in this report deliberately trades some accuracy for substantially higher recall/precision on the clinically important minority class.

4. Modeling Methodology

Split: stratified 60% train / 20% validation / 20% test, random_state=42, identical across all six models so comparisons are fair. The test set was touched exactly once per model, for final evaluation only. Hyperparameter tuning used RandomizedSearchCV with StratifiedKFold (3 folds) scoring on average_precision for the four tree-based models; the two neural networks (LSTM, MLP) instead used a compact architecture search selected by held-out validation PR-AUC (k-fold CV was judged too computationally expensive to repeat per candidate on this machine's 8 GB RAM budget). In every case the baseline-vs-tuned decision was made from CV/validation performance only, never from the test set.

5. Majority-Class Baseline

Accuracy=0.7766, Precision=0, Recall=0, F1=0, PR-AUC≈0.2234. Every tuned model in this report substantially exceeds this PR-AUC floor.

6. Per-Model Results (Baseline vs Tuned)

Decision Tree

Selection basis: 0.330672 (baseline) vs 0.332507 (search) → **Tuned** retained. Final test: PR-AUC=0.3320, Precision=0.2878, Recall=0.7234, F1=0.4118, ROC-AUC=0.6473, Accuracy=0.5385, Threshold=0.2159. Generalization: Train PR-AUC=0.3351 / Test PR-AUC=0.3320 (gap 0.0031, Well-Balanced). Execution time: 26.2s.

Hyperparameters: {'class_weight': None, 'criterion': 'gini', 'max_depth': 6, 'max_features': None, 'min_samples_leaf': 110, 'min_samples_split': 150}.

Random Forest

Selection basis: 0.343166 (baseline) vs 0.343399 (search) → **Tuned** retained. Final test: PR-AUC=0.3429, Precision=0.2986, Recall=0.6852, F1=0.4160, ROC-AUC=0.6526, Accuracy=0.5702, Threshold=0.4732. Generalization: Train PR-AUC=0.3914 / Test PR-AUC=0.3429 (gap 0.0484, Moderate Overfit). Execution time: 113.4s. Hyperparameters: {'class_weight': 'balanced_subsample', 'criterion': 'gini', 'max_depth': 15, 'max_features': 'log2', 'min_samples_leaf': 68, 'min_samples_split': 78, 'n_estimators': 91}.

XGBoost

Selection basis: 0.345894 (baseline) vs 0.347908 (search) → **Tuned** retained. Final test: PR-AUC=0.3465, Precision=0.3025, Recall=0.6683, F1=0.4165, ROC-AUC=0.6545, Accuracy=0.5817, Threshold=0.4841. Generalization: Train PR-AUC=0.3499 / Test PR-AUC=0.3465 (gap 0.0034, Well-Balanced). Execution time: 102.1s. Hyperparameters: {'colsample_bytree': 0.8, 'gamma': 0.5, 'learning_rate': 0.03, 'max_depth': 3, 'min_child_weight': 8, 'reg_alpha': 0.01, 'reg_lambda': 0.5, 'scale_pos_weight': 3.477, 'subsample': 0.9}.

Gradient Boosting

Selection basis: 0.346952 (baseline) vs 0.347040 (search) → **Tuned** retained. Final test: PR-AUC=0.3464, Precision=0.3043, Recall=0.6609, F1=0.4168, ROC-AUC=0.6545, Accuracy=0.5868, Threshold=0.4912. Generalization: Train PR-AUC=0.3535 / Test PR-AUC=0.3464 (gap 0.0071, Well-Balanced). Execution time: 32.0s. Hyperparameters: {'class_weight': 'balanced', 'l2_regularization': 0.5, 'learning_rate': 0.05, 'max_bins': 255, 'max_depth': 7, 'max_iter': 287, 'max_leaf_nodes': 30, 'min_samples_leaf': 29}.

LSTM

Selection basis: 0.348685 (baseline) vs 0.348685 (search) → **Baseline** retained. Final test: PR-AUC=0.3452, Precision=0.2990, Recall=0.6842, F1=0.4162, ROC-AUC=0.6540, Accuracy=0.5712, Threshold=0.4794. Generalization: Train PR-AUC=0.3503 / Test PR-AUC=0.3452 (gap 0.0051, Well-Balanced). Execution time: 2.5s. Hyperparameters: {'lstm_units': 64, 'dense_units': 32, 'dropout': 0.3, 'learning_rate': 0.001, 'batch_size': 2048}.

Sequence-structure limitation: 97.0% of unique patients have exactly one row in this dataset (only ~2.95% have repeat visits, max 4). No genuine multi-step temporal sequence exists, so seq_len=1 was used deliberately rather than fabricating artificial sequences. Note: an initial LSTM run was degenerate (PR-AUC≈0.222, ROC-AUC≈0.50) due to missing input feature scaling, which saturated the recurrent gates; this was diagnosed and fixed by adding a train-only-fitted StandardScaler.

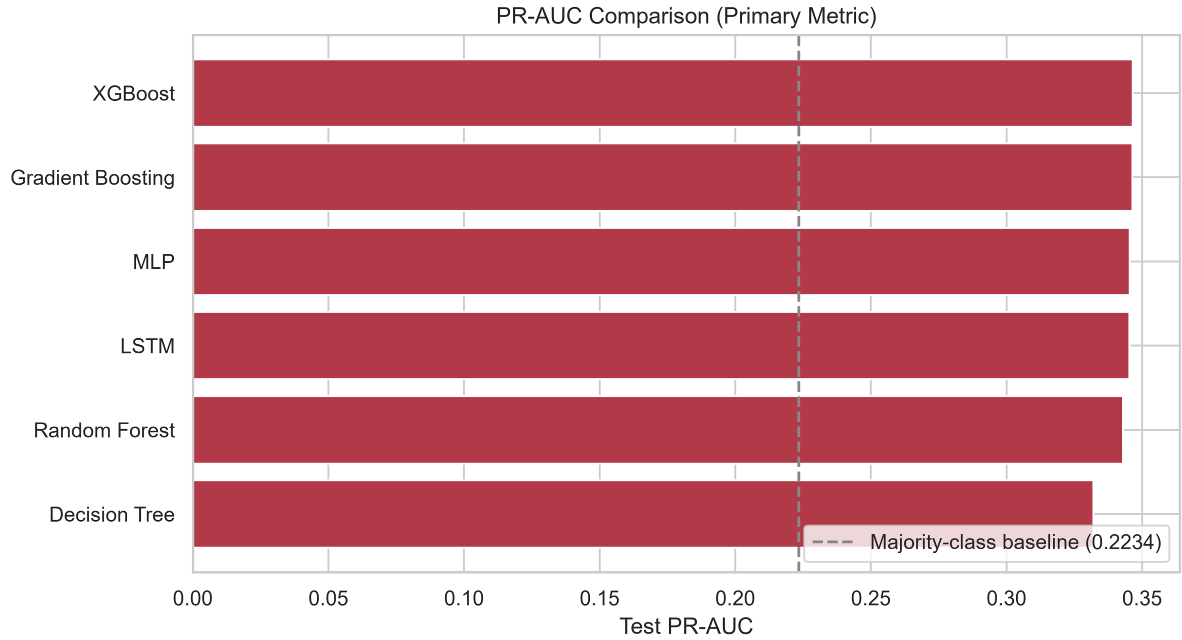
MLP

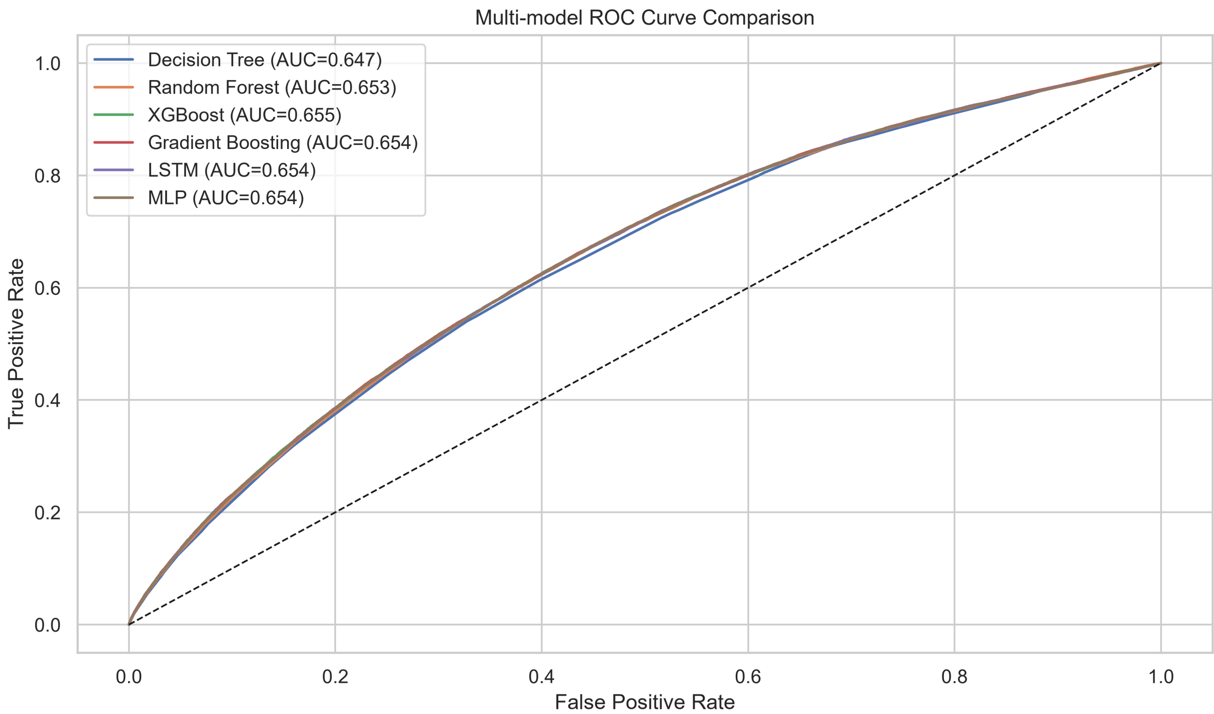
Selection basis: 0.347670 (baseline) vs 0.348918 (search) → **Tuned (deeper)** retained. Final test: PR-AUC=0.3454, Precision=0.2946, Recall=0.7158, F1=0.4174, ROC-AUC=0.6540, Accuracy=0.5537, Threshold=0.4660. Generalization: Train PR-AUC=0.3500 / Test PR-AUC=0.3454 (gap 0.0046, Well-Balanced). Execution time: 2.5s. Hyperparameters: {'hidden_units': [128, 64, 32], 'dropout': [0.3, 0.3, 0.2], 'l2': 0.0001, 'learning_rate': 0.001, 'batch_size': 2048}.

7. Threshold Optimization Methodology

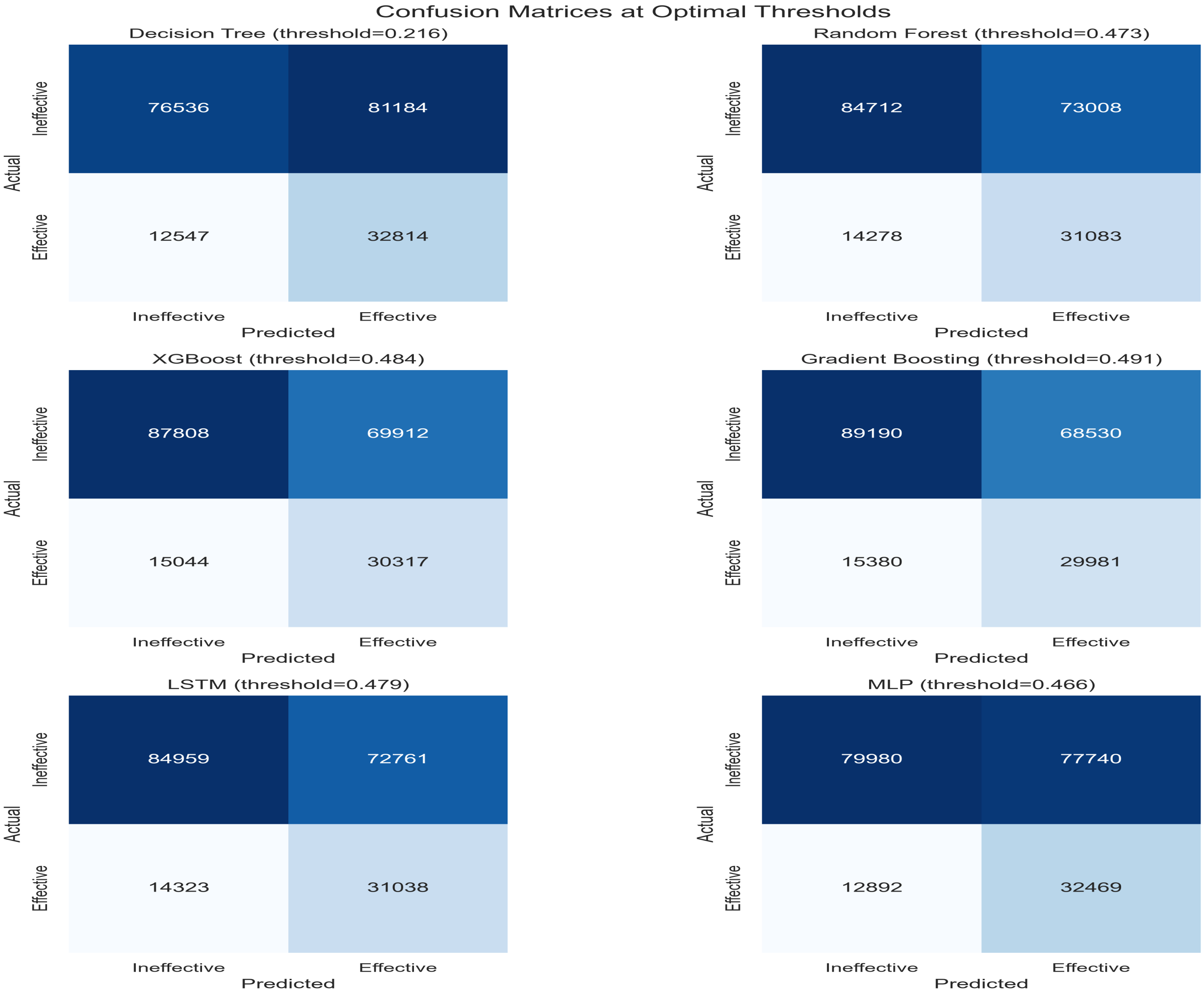
For every model, probabilities were generated on an inner validation split never seen by the fitted estimator's parameters, and the decision threshold was chosen by maximizing F1 across the precision-recall curve on that validation split (find_best_threshold in utils.py). The chosen threshold was then applied exactly once to the untouched test set. The test set was never used to choose a threshold.

8. Precision-Recall and ROC-AUC Analysis

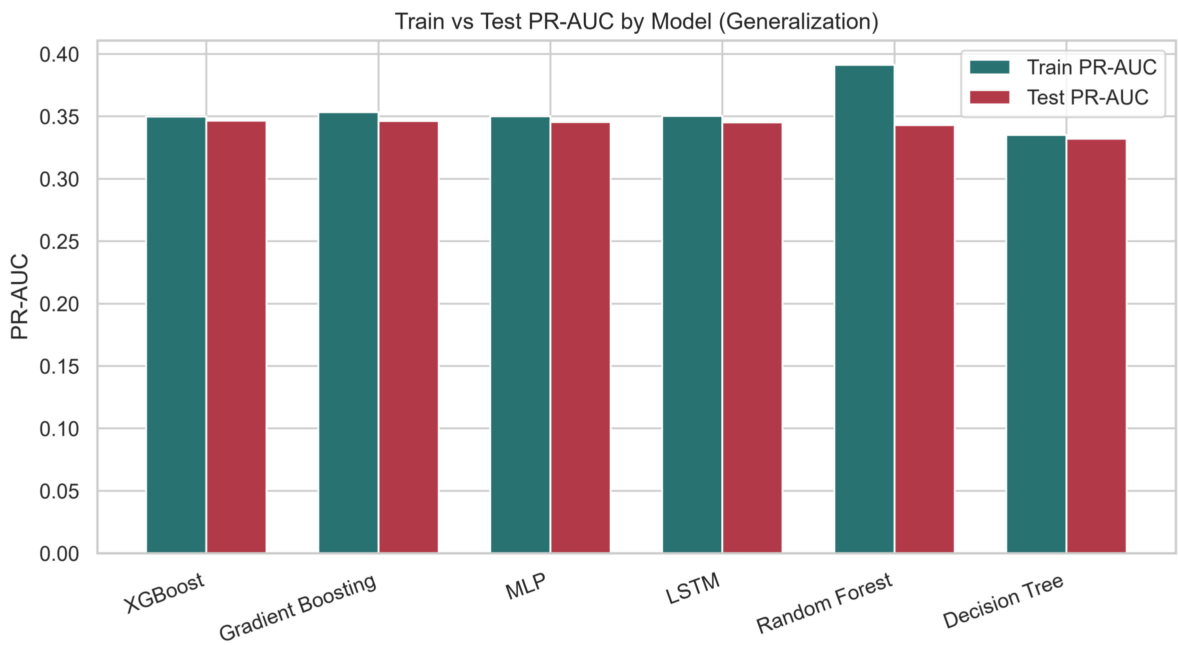




9. Confusion Matrix Comparison


Confusion Matrices at Optimal Thresholds
Decision Tree (threshold=0.216)
76536
81184
12547
32814
Random Forest (threshold=0.473)
84712
73008
14278
31083
XGBoost (threshold=0.484)
87808
69912
15044
30317
Gradient Boosting (threshold=0.491)
89190
68530
15380
29981
LSTM (threshold=0.479)
84959
72761
14323
31038
MLP (threshold=0.466)
79980
77740
12892
32469
Actual
Ineffective
Effective
Predicted
Ineffective
Effective

10. Overfitting and Generalization Comparison



Model	Train PR-AUC	Test PR-AUC	PR-AUC Gap	Train F1	Test F1	F1 Gap	Diagnosis
XGBoost	0.3499	0.3465	0.0034	0.4193	0.4165	0.0029	Well-Balanced
Gradient Boosting	0.3535	0.3464	0.0071	0.4209	0.4168	0.0041	Well-Balanced
MLP	0.3500	0.3454	0.0046	0.4206	0.4174	0.0031	Well-Balanced
LSTM	0.3503	0.3452	0.0051	0.4207	0.4162	0.0046	Well-Balanced
Random Forest	0.3914	0.3429	0.0484	0.4410	0.4160	0.0250	Moderate Overfit
Decision Tree	0.3351	0.3320	0.0031	0.4159	0.4118	0.0041	Well-Balanced

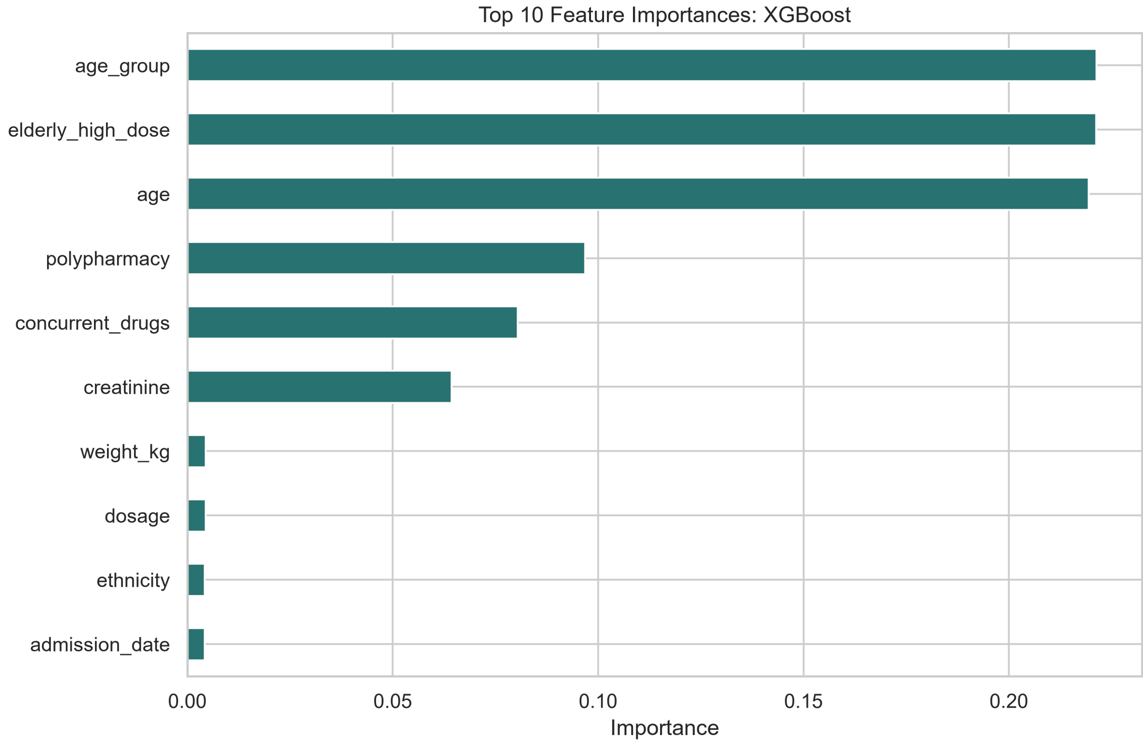
11. Final Model Comparison Table

Rank	Model	PR-AUC	ROC-AUC	Precision	Recall	F1	Accuracy	Threshold	Diagnosis
1	XGBoost	0.3465	0.6545	0.3025	0.6683	0.4165	0.5817	0.4841	Well-Balanced
2	Gradient Boosting	0.3464	0.6545	0.3043	0.6609	0.4168	0.5868	0.4912	Well-Balanced
3	MLP	0.3454	0.6540	0.2946	0.7158	0.4174	0.5537	0.4660	Well-Balanced
4	LSTM	0.3452	0.6540	0.2990	0.6842	0.4162	0.5712	0.4794	Well-Balanced
5	Random Forest	0.3429	0.6526	0.2986	0.6852	0.4160	0.5702	0.4732	Moderate Overfit
6	Decision Tree	0.3320	0.6473	0.2878	0.7234	0.4118	0.5385	0.2159	Well-Balanced

12. Champion Model Selection

Champion: Gradient Boosting. XGBoost and Gradient Boosting are statistically tied on PR-AUC (margin 0.000185, likely within run-to-run noise). Gradient Boosting is selected as champion because it has an equal-or-better F1 (0.4168 vs 0.4165) and comparable generalization (Well-Balanced), giving a slightly better precision-recall balance for identifying Effective cases. Selection deliberately does NOT default to the highest accuracy, highest recall, highest ROC-AUC, or highest training score alone; PR-AUC, Class-1 precision/recall balance, F1, generalization gap, and computational practicality were all weighed.

13. Feature Importance



14. Limitations

(1) All six models cluster within a narrow PR-AUC band (~0.33-0.35), consistent with weak individual feature signal observed during data exploration (maximum mutual-information score ≈0.011 for 'age'); this likely reflects a ceiling in the available features rather than a modeling deficiency. (2) The LSTM has no genuine sequential structure to exploit (97% single-record patients); its recurrent architecture is not meaningfully advantaged over the MLP here. (3) Neural-network hyperparameter selection used a single validation split rather than k-fold CV, for computational reasons on an 8 GB RAM machine; tree-model tuning used only 3-fold CV and a compact search budget for the same reason, so the explored hyperparameter space, while reasonable, was not exhaustive. (4) A subset of raw columns (e.g. adverse_event, readmission_30d) have unclear causal timing relative to the target and were retained as-is; genuine leakage was not confirmed but cannot be fully ruled out from column names alone.

15. Recommendations for Future Work

Investigate richer feature engineering (interaction terms, non-linear transforms of lab values) given the weak individual mutual-information scores; obtain genuinely longitudinal per-patient records if repeat-visit sequences become available, to make an LSTM/sequence model worthwhile; consider probability calibration (e.g., Platt scaling/isotonic regression) before any clinical use; run a wider hyperparameter search with more compute if available; and perform subgroup/fairness analysis across demographic segments before deployment.

16. Reproducibility

random_state=42 throughout (data split, model fitting, CV folds). Split: 60/20/20 stratified on treatment_outcome. Preprocessing: FrequencyCategoryEncoder (median/frequency-rank, train-only fit), plus StandardScaler (train-only fit) for LSTM/MLP. Threshold method: F1-maximization on validation predictions. CV strategy: StratifiedKFold(3) for tree models scoring average_precision; held-out validation PR-AUC for neural networks. All hyperparameters, execution times, and model versions are recorded in metrics_output/*_comparison.json and *_metrics.json / *.json alongside this report.